

| Question          |                                |                                 |    | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks available |                  |                                 |       |                         |               |                                |       |   |      |               |                      |        |   |        |                   |                     |        |    |        |       |                  |       |   |         |                  |                   |      |    |        |  |   |  |   |  |   |
|-------------------|--------------------------------|---------------------------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------|---------------------------------|-------|-------------------------|---------------|--------------------------------|-------|---|------|---------------|----------------------|--------|---|--------|-------------------|---------------------|--------|----|--------|-------|------------------|-------|---|---------|------------------|-------------------|------|----|--------|--|---|--|---|--|---|
|                   |                                |                                 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | AO1             | AO2              | AO3                             | Total | Maths                   | Prac          |                                |       |   |      |               |                      |        |   |        |                   |                     |        |    |        |       |                  |       |   |         |                  |                   |      |    |        |  |   |  |   |  |   |
| 2                 | (a)                            | (i)                             |    | correction 1 – (should be) acid (1)<br>correction 2 – (should be pH) 7 (1)<br>correction 3 – (should be) Na <sub>2</sub> CO <sub>3</sub> (1)<br><br>award credit for correct corrections in any order                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1<br>1          | 1                |                                 | 3     |                         | 3             |                                |       |   |      |               |                      |        |   |        |                   |                     |        |    |        |       |                  |       |   |         |                  |                   |      |    |        |  |   |  |   |  |   |
|                   |                                | (ii)                            |    | <table border="1"><thead><tr><th>Chemical Name</th><th>Chemical formula</th><th>Colour with Universal Indicator</th><th>pH</th><th>Acid, Alkali or Neutral</th></tr></thead><tbody><tr><td>Sulfuric acid</td><td>H<sub>2</sub>SO<sub>4</sub></td><td>Green</td><td>1</td><td>Acid</td></tr><tr><td>Ethanoic acid</td><td>CH<sub>3</sub>COOH</td><td>Orange</td><td>4</td><td>Alkali</td></tr><tr><td>Calcium hydroxide</td><td>Ca(OH)<sub>2</sub></td><td>Purple</td><td>14</td><td>Alkali</td></tr><tr><td>Water</td><td>H<sub>2</sub>O</td><td>Green</td><td>5</td><td>Neutral</td></tr><tr><td>Sodium carbonate</td><td>NaCO<sub>3</sub></td><td>Blue</td><td>10</td><td>Alkali</td></tr></tbody></table> <p>error 1</p> <p>error 2</p> <p>error 3</p> <p>'green' circled</p> <p>do not accept more than one circled</p> | Chemical Name   | Chemical formula | Colour with Universal Indicator | pH    | Acid, Alkali or Neutral | Sulfuric acid | H <sub>2</sub> SO <sub>4</sub> | Green | 1 | Acid | Ethanoic acid | CH <sub>3</sub> COOH | Orange | 4 | Alkali | Calcium hydroxide | Ca(OH) <sub>2</sub> | Purple | 14 | Alkali | Water | H <sub>2</sub> O | Green | 5 | Neutral | Sodium carbonate | NaCO <sub>3</sub> | Blue | 10 | Alkali |  | 1 |  | 1 |  | 1 |
| Chemical Name     | Chemical formula               | Colour with Universal Indicator | pH | Acid, Alkali or Neutral                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 |                  |                                 |       |                         |               |                                |       |   |      |               |                      |        |   |        |                   |                     |        |    |        |       |                  |       |   |         |                  |                   |      |    |        |  |   |  |   |  |   |
| Sulfuric acid     | H <sub>2</sub> SO <sub>4</sub> | Green                           | 1  | Acid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                 |                  |                                 |       |                         |               |                                |       |   |      |               |                      |        |   |        |                   |                     |        |    |        |       |                  |       |   |         |                  |                   |      |    |        |  |   |  |   |  |   |
| Ethanoic acid     | CH <sub>3</sub> COOH           | Orange                          | 4  | Alkali                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                 |                  |                                 |       |                         |               |                                |       |   |      |               |                      |        |   |        |                   |                     |        |    |        |       |                  |       |   |         |                  |                   |      |    |        |  |   |  |   |  |   |
| Calcium hydroxide | Ca(OH) <sub>2</sub>            | Purple                          | 14 | Alkali                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                 |                  |                                 |       |                         |               |                                |       |   |      |               |                      |        |   |        |                   |                     |        |    |        |       |                  |       |   |         |                  |                   |      |    |        |  |   |  |   |  |   |
| Water             | H <sub>2</sub> O               | Green                           | 5  | Neutral                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 |                  |                                 |       |                         |               |                                |       |   |      |               |                      |        |   |        |                   |                     |        |    |        |       |                  |       |   |         |                  |                   |      |    |        |  |   |  |   |  |   |
| Sodium carbonate  | NaCO <sub>3</sub>              | Blue                            | 10 | Alkali                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                 |                  |                                 |       |                         |               |                                |       |   |      |               |                      |        |   |        |                   |                     |        |    |        |       |                  |       |   |         |                  |                   |      |    |        |  |   |  |   |  |   |
|                   | (b)                            | (i)                             |    | carbon dioxide / CO <sub>2</sub> (1)<br><br>barium chloride / BaCl <sub>2</sub> (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2               |                  |                                 | 2     |                         | 2             |                                |       |   |      |               |                      |        |   |        |                   |                     |        |    |        |       |                  |       |   |         |                  |                   |      |    |        |  |   |  |   |  |   |
|                   |                                | (ii)                            |    | <u>limewater</u> goes <u>milky</u><br><br>accept alternative descriptions to milky e.g. turns white / cloudy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1               |                  |                                 | 1     |                         | 1             |                                |       |   |      |               |                      |        |   |        |                   |                     |        |    |        |       |                  |       |   |         |                  |                   |      |    |        |  |   |  |   |  |   |